

# DATA ANALYST ROADMAP — INDIA → WORLD

Complete Data Analyst Bootcamp: Basics to Advanced (Udemy · Krish Naik & Jayant Topnani · 89 hrs · 4.6★) + Post-Course Skill Tracks

**Course pace:** 1.25× speed · 2 hrs content/day + 1 hr practice/notes **Goal 1:** Finish course in ~7 weeks

**Goal 2:** India DA job within 3 months of course end **Goal 3:** Common global skills

**Goal 4:** Europe + Australia dual tracks

**89 hrs**

Course at 1× speed

**1 hr/day**

Notes + code-along practice

**~71 hrs**

Effective at 1.25× speed

**~36 days**

Pure content days @ 2hr/day

**2 hrs/day**

Content watched (1.6 real hrs)

**7 weeks**

Realistic finish (with buffer)

## ⚡ DAILY NON-NEGOTIABLES — DO THESE EVERY SINGLE DAY

- |                                                                                                |                                                                                              |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Watch 2 hrs course content at 1.25× speed (= ~1.6 hrs real time)      | <input type="checkbox"/> 1 hr: code along + take notes in your own words (not copy-paste)    |
| <input type="checkbox"/> Push at least one file/script to GitHub (even 5 lines = habit)        | <input type="checkbox"/> Write one line in your learning log: what did you understand today? |
| <input type="checkbox"/> After Week 5 (SQL starts): 2 LeetCode SQL problems/day, Easy → Medium | <input type="checkbox"/> Never skip 2 days in a row — momentum is everything                 |

⚠ **ACTUAL COURSE ORDER (verified from Udemy curriculum screenshots):** Python Basics → NumPy/Pandas/Matplotlib/Seaborn → Statistics (Descriptive + Probability Distributions + Inferential/Hypothesis Testing) → Feature Engineering → EDA Projects (Red Wine, Flight Price, Google Playstore ×2) → SQL (Basics → Advanced → Interview Qs) → Power BI + DAX (3 projects) → Excel (2 dashboards) → Tableau (2 dashboards + SQL+Tableau project) → Snowflake → Cloud End-to-End Projects (Dataflow, MySQL, BigQuery, Azure) → AI Power BI Project → GitHub → Prompt Engineering & Gen AI

## ■ PHASE 1 — COURSE SPRINT

Weeks 1-7 · 3 hrs/day · Follow course in exact order

### WEEK 1 — Setup + Python Foundations

Day 1-7 · Sections: Introduction · Getting Started With Python · Complete Python Part 1 · ~12 hrs

#### ★ DAY 1 — BEFORE ANY COURSE CONTENT

- ☐ Install Git. Create GitHub account. Learn 5 commands: init, add, commit, push, clone
- ☐ Create repo data-analyst-portfolio — every project lives here from Day 1
- ☐ Learn Markdown basics (1 hr) — you'll write README for every project
- ☐ Install Anaconda + VS Code. Verify Google Colab works (course uses both)
- ☐ Watch course intro: "What Does A Data Analyst Do and Its Roadmap" (18 min)

#### ■ GETTING STARTED WITH PYTHON (2 LECTURES · 19 MIN)

- ☐ Getting Started With Google Colab (7 min)
- ☐ Installation Of Anaconda And Visual Studio Code (11 min)

#### ■ COMPLETE PYTHON WITH IMPORTANT LIBRARIES — PART 1 (DAYS 2-7) · 29 LECTURES · 7HR 3MIN

- ☐ Getting Started With VS Code With Environments (12 min)
- ☐ **Python Basics: Syntax And Semantics** (20 min)
- ☐ **Variables In Python** (18 min)
- ☐ **Basic Data Types In Python** — int, float, str, bool (9 min)
- ☐ **Operators In Python** (16 min) + Coding Exercise And Assignments

- ☐ **Conditional Statements (if/elif/else)** (21 min) + Coding Exercise And Assignments
- ☐ **Loops In Python** – for, while, break, continue (28 min) + Coding Exercise And Assignments
- ☐ **List And List Comprehension In Python** (37 min) + List Practise Code And Assignments
- ☐ **Tuples In Python** (22 min) + Tuple Assignment And Practise Code
- ☐ **Sets In Python** (21 min) + Sets Assignment and Practise Code
- ☐ **Dictionaries In Python** (38 min) + Dictionaries Assignments And Practise Questions

💡 Push Python practice scripts to GitHub every day from Day 2 – even 5 lines counts. Course exercise files = daily commits.

## WEEK 2 – Python Complete + NumPy + Pandas + Matplotlib + Seaborn

Day 8-14 · Sections: Complete Python Part 2 · Data Analysis With Python (7 lectures · 2hr 27min)

### ■ COMPLETE PYTHON – PART 2 (DAYS 8-10)

- ☐ **Real World Use Cases of List** (9 min)
- ☐ **Getting Started With Functions** (24 min)
- ☐ **More Coding Examples With Functions** (28 min)
- ☐ **Lambda Functions** (9 min)
- ☐ **Map Functions In Python** (11 min)
- ☐ **Filter Function In Python** (9 min) + Function Assignments
- ☐ **Import Modules And Packages In Python** (17 min)
- ☐ **Standard Library Overview** (17 min)
- ☐ **File Operation In Python** (17 min) + Working With File Paths (8 min)
- ☐ **Exception Handling: Try/Except/else/finally** (25 min)

### ■ DATA ANALYSIS WITH PYTHON – 7 LECTURES · 2HR 27MIN (DAYS 11-14)

- ☐ **Numpy In Python** (28 min) – arrays, shape, dtype, broadcasting, vectorised ops
- ☐ **Pandas: DataFrame And Series** (29 min) – create, index, slice, dtypes
- ☐ **Data Manipulation With Pandas And Numpy** (24 min) – sort, filter, groupby, merge, concat, apply, pivot\_table
- ☐ **Numpy Assignments With Solution** – complete all
- ☐ **Reading Data From Various Data Source Using Pandas** (15 min) – CSV, Excel, SQL, web
- ☐ **Data Visualization With Matplotlib** (30 min) – line, bar, scatter, histogram, pie, subplots, labels, figsize
- ☐ **Data Visualization With Seaborn** (18 min) – heatmap, boxplot, violin, pairplot, countplot, regplot

💡 Matplotlib + Seaborn come immediately after Pandas in the course. Practice: load any CSV, clean it, produce 5 different charts. Push notebook to GitHub.

## ■ PHASE 1 CONTINUED – Course Sprint

Weeks 3-4 · Statistics is 3 full sections (~9 hrs total). Don't underestimate it.

### WEEK 3 – Statistics: All Three Sections (Descriptive + Probability + Inferential)

Day 15-21 · 6 + 8 + 13 + 15 lectures = ~9 hrs of stats. Take your time. These questions come up in interviews.

#### ■ GETTING STARTED WITH STATISTICS – 6 LECTURES · 1HR 33MIN

- ☐ Introduction To Statistics (10 min)
- ☐ Types Of Statistics (12 min) – descriptive vs inferential
- ☐ Population And Sample Data (15 min)
- ☐ Types Of Sampling Techniques (23 min)
- ☐ Types Of Data (8 min) – nominal, ordinal, interval, ratio
- ☐ Scales Of Measurement Of Data (22 min)

#### ■ DESCRIPTIVE STATISTICS – 8 LECTURES · 2HR 30MIN

- ☐ Measure Of Central Tendency: Mean, Median, Mode (25 min)
- ☐ Measures Of Dispersion: Range, Variance, Standard Deviation (23 min)
- ☐ Why Sample Variance is divided by n-1 (8 min)
- ☐ Random Variables (8 min)
- ☐ Percentiles And Quartiles (10 min)
- ☐ 5-Number Summary (18 min)
- ☐ Histogram And Skewness (25 min)
- ☐ Covariance And Correlation (39 min)

#### ■ PROBABILITY DISTRIBUTION FUNCTION AND TYPES – 13 LECTURES · 3HR

- ☐ PDF, PMF, CDF (31 min)
- ☐ Types Of Probability Distribution (8 min)
- ☐ Bernoulli Distribution (24 min)
- ☐ Binomial Distribution (10 min)
- ☐ Poisson Distribution (13 min)
- ☐ Normal / Gaussian Distribution (14 min)
- ☐ Standard Normal Distribution (14 min)
- ☐ Uniform + Log Normal + Power Law + Pareto distributions
- ☐ Central Limit Theorem (10 min) + Estimators (6 min)

#### ■ INFERENCE STATISTICS AND HYPOTHESIS TESTING – 15 LECTURES · 2HR 31MIN

- ☐ Hypothesis Testing And Mechanism (7 min)
- ☐ P value And Hypothesis Testing (12 min)
- ☐ **SKIM** Z test Hypothesis Testing (27 min) – concept only, 10 min
- ☐ Student t Distribution (5 min) + T stats With T test (10 min)
- ☐ Z test vs T test (3 min) – know when to use which
- ☐ Type 1 And Type 2 Error (5 min) + Bayes Theorem (9 min)
- ☐ Confidence Interval And Margin Of Error (8 min)
- ☐ **SKIM** Chi Square Test (8+12 min) – concept only
- ☐ **SKIM** ANOVA: What Is + Assumptions + Types + Partitioning – 15 min total

🕒 Z-test, T-test, ANOVA, Chi-Square: concept only (10-15 min each). Not hand-calculated in any DA interview. Know what they are and when to use them.

**WEEK 4 – Feature Engineering + All 4 EDA Course Projects** Day 22-28 · Sections: Feature Engineering With Python (7 lectures · 1hr 24min) · Exploratory Data Analysis (4 lectures · 1hr 39min)

**FEATURE ENGINEERING WITH PYTHON – 7 LECTURES · 1HR 24MIN**

- |                                                                                                               |                                                                                               |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> <b>Handling Missing Data</b> (20 min) – mean/median/mode imputation, drop strategies | <input type="checkbox"/> <b>Handling Outliers With Python</b> (8 min) – IQR, Z-score, capping |
| <input type="checkbox"/> <b>Handling Imbalanced Dataset</b> (12 min)                                          | <input type="checkbox"/> <b>Data Encoding: Nominal / One Hot Encoding</b> (13 min)            |
| <input type="checkbox"/> <b>Feature Engineering: SMOTE</b> (13 min) – synthetic oversampling                  | <input type="checkbox"/> <b>Label And Ordinal Encoding</b> (9 min)                            |
|                                                                                                               | <input type="checkbox"/> <b>Target Guided Ordinal Encoding</b> (5 min)                        |

**EXPLORATORY DATA ANALYSIS – 4 COURSE PROJECTS · 4 LECTURES · 1HR 39MIN · PUSH ALL TO GITHUB**

**PROJECT A – Red Wine Dataset EDA (23 min)**

*Skills: Pandas · NumPy · Matplotlib · Seaborn · Feature Engineering · Descriptive Statistics*

- ☐ Load data, check shape/dtypes/nulls, describe()
- ☐ Distribution plots, correlation heatmap, outlier detection
- ☐ Write 5 findings in Jupyter markdown cells
- ☐ Push notebook + README to GitHub

**PROJECT C – Data Cleaning Google Playstore – Part 1 (26 min)**

*Skills: Pandas (messy string cleaning) · dtype conversion · handling corrupt rows*

- ☐ Fix Ratings (non-numeric), Size (MB/KB strings), Installs (commas/+)
- ☐ Fix Price (\$ signs), clean Type, handle missing values
- ☐ Save cleaned CSV – input for Part 2. Push to GitHub.

**PROJECT B – EDA Flight Price Dataset (30 min)**

*Skills: Pandas · Feature Engineering (date parsing, encoding) · Seaborn · Matplotlib*

- ☐ Clean: handle missing, parse duration/dates into numeric
- ☐ Engineer: duration in minutes, day of week, peak/off-peak
- ☐ Visualise: price by airline, stops, route. Push to GitHub.

**PROJECT D – EDA Google Play Store – Part 2 (18 min)**

*Skills: Matplotlib · Seaborn · groupby analysis · business interpretation*

- ☐ Top categories, rating distributions, free vs paid analysis
- ☐ Reviews vs installs correlation, content rating breakdown
- ☐ Write conclusions in Jupyter markdown. Push both parts to GitHub.

💡 These 4 EDA projects complete the Python/Stats half of the course. By end of Week 4 you can clean, analyse, and visualise real data. SQL starts next week.

## ■ PHASE 1 CONTINUED — Weeks 5-7 · SQL comes AFTER EDA in the actual course. Power BI projects are Sales, Insurance, UPI — not Loan/Bank/Real Estate/HR.

### Course Sprint

**WEEK 5 — SQL: Basics** → Day 29-35 · SQL Intro (25min) · MS SQL Server Basics (47 lec · 9hr 3min) · SQL Basics Advanced → Interview Qs (1hr 27min) · SQL Assignments · SQL Functions (1hr 26min) · Advanced SQL (1hr 39min) · SQL Interview Qs (57min)

**Questions**

#### ■ SQL FOUNDATIONS + ALL ASSIGNMENTS

- ☐ SQL Course Introduction & Overview (3 lectures · 25 min)
- ☐ Microsoft SQL Server Basics (47 lectures · 9hr 3min) — install SQL Server Express, run every query yourself
- ☐ SELECT, DISTINCT, WHERE, ORDER BY, AND/OR/NOT, BETWEEN, IN, NULL handling
- ☐ INSERT INTO, UPDATE, DELETE, DROP, TRUNCATE
- ☐ MAX/MIN/SUM/AVG/COUNT + GROUP BY + HAVING
- ☐ ALL JOINS: INNER, LEFT, RIGHT, FULL OUTER, SELF, ANTI, UNION / UNION ALL
- ☐ Subqueries (correlated vs uncorrelated), CASE statements, Nested CASE
- ☐ Data Types, CREATE TABLE, ALTER TABLE, Constraints (PK, FK, NOT NULL, UNIQUE, CHECK)
- ☐ SQL Order of Execution: FROM→WHERE→GROUP BY→HAVING→SELECT→ORDER BY→LIMIT
- ☐ ☒ SQL Basics Questions (4 lectures · 1hr 27min) — complete all
- ☐ ☒ SQL Assignments — complete EVERY assignment (no skipping)
- ☐ ☒ SQL Functions (7 lectures · 1hr 26min) — complete all

🚫 Do every SQL assignment. Watching without doing = 0 interview SQL. Start LeetCode SQL from Day 29: 2 problems/day, Easy first.

#### ■ ADVANCED SQL + INTERVIEW QUESTIONS

- ☐ **Advanced SQL (8 lectures · 1hr 39min):**
- ☐ Window Functions: RANK, DENSE\_RANK, ROW\_NUMBER, LEAD, LAG, FIRST\_VALUE, LAST\_VALUE
- ☐ CTEs (WITH clause, chained CTEs) + Recursive CTEs
- ☐ Stored Procedures (create + execute + params)
- ☐ Views (create, query, update, drop)
- ☐ Indexes: Clustered vs Non-Clustered
- ☐ ISNULL, COALESCE — null handling in real queries
- ☐ **SQL Important Interview Questions (4 lectures · 57min):**
- ☐ Nth Highest Salary query
- ☐ Reportee/Manager hierarchy query
- ☐ Deleting Duplicates — Method 1 + Method 2

💡 Window functions + CTEs are what separate junior from mid-level in every SQL interview. Don't rush these 8 lectures.

## WEEK 6 – Power BI + DAX + Excel + Tableau + SQL+Tableau Project

Day 36-45 · Allow 9-10 days. Power BI has 3 projects (Sales, Insurance, UPI). Excel has 2 dashboards. Tableau has 2 dashboards + SQL+Tableau project.

### ■ POWER BI – COMPLETE ALL SECTIONS

- ☐ Power BI Course Introduction (1 lec · 11 min) + Introduction to Power BI (4 lec · 22 min)
- ☐ **Data Visualization** (15 lec · 2hr 22min) – all chart types, KPI, card, matrix, map, treemap, ribbon, funnel, gauge
- ☐ **Power Query Editor** (12 lec · 2hr 12min) – all transforms, append, merge, group by, pivot, unpivot
- ☐ **DAX (8 lec · 2hr 26min)** – Calculated Columns vs Measures · CALCULATE+FILTER · Time Intelligence (TOTALYTD, SAMEPERIODLASTYEAR) · RANKX, TOPN · DIVIDE, IF, SWITCH · ALL, ALLEXCEPT, RELATED

💡 DAX is the most important section in all of Power BI. Take notes by hand. Recreate every formula from scratch before moving on.

### 🔧 POWER BI COURSE PROJECTS – ALL 3 (CONFIRMED FROM CURRICULUM)

- ☐ **Project 1: Sales Data Analysis** (16 lec · 3hr 39min) – covers Data Modeling. Skills: Power Query, DAX, data model, relationships, slicers, conditional formatting
- ☐ **Project 2: Insurance Data Analysis** (18 lec · 3hr 22min) – Skills: advanced DAX, drill-through, RLS, row-level security
- ☐ **Project 3: UPI Transactions Data Analysis** (11 lec · 1hr 30min) – Skills: time intelligence, KPI cards, publishing to Power BI Service
- ☐ Miscellaneous Section Power BI (2 lec · 23min) – publishing, scheduling refresh, RLS

### ■ MICROSOFT EXCEL – COMPLETE ALL SECTIONS

- ☐ **Getting Started with Microsoft Excel** (17 lec · 2hr 42min) – formatting, VLOOKUP, XLOOKUP, pivot tables, conditional formatting
- ☐ **Excel Dashboard 1** (8 lec · 1hr 25min) – full dashboard, push to GitHub
- ☐ **Excel Dashboard 2** (8 lec · 1hr 22min) – second complete dashboard
- ☐ **Power Query Editor (MS Excel)** (21 lec · 2hr 5min)
- ☐ **Excel Activity: Importing Data From SQL Server** (6 lec · 35min) – connects Excel to SQL Server directly

### ■ TABLEAU – COMPLETE ALL SECTIONS

- ☐ **Tableau** (21 lec · 3hr 40min) – all chart types, filters, sets, parameters, maps, LOD intro, dashboards
- ☐ **Tableau Dashboard 1** (6 lec · 36min) – publish to Tableau Public (free, get public URL)
- ☐ **Tableau Dashboard 2** (6 lec · 35min)
- ☐ **Tableau Prep Builder** (12 lec · 1hr 37min) – full ETL prep workflow
- ☐ **SQL + Tableau Project: Student Depression Data Analysis** (14 lec · 1hr 27min) – SQL queries + Tableau dashboard on same dataset. Write GitHub README.

💡 The SQL+Tableau project is a course-built end-to-end project. Treat it like a portfolio piece – proper README, screenshots, published Tableau Public link.

## WEEK 7 – Snowflake + Cloud End-to-End Projects + AI Power BI + GitHub + Prompt Engineering

Day 46-52 · Final course sections. 4 cloud end-to-end Power BI projects. These replicate real enterprise data stacks.

### ■ SNOWFLAKE + CLOUD INTEGRATIONS

- ☐ **Snowflake** (15 lec · 1hr 9min) – architecture, virtual warehouses, SQL worksheets, stages, data loading
- ☐ **Connecting Snowflake to Power BI & Tableau** (2 lec · 9min)
- ☐ **AWS + Snowflake + Power BI Project** (11 lec · 1hr 10min) – S3 → Snowflake → Power BI full pipeline
- ☐ **AWS + Snowflake + Tableau Project** (11 lec · 1hr 6min) – S3 → Snowflake → Tableau full pipeline

### ■ END-TO-END POWER BI PROJECTS (4 PROJECTS – DIFFERENT DATA SOURCES)

- ☐ **Project 1: Datasource – Dataflow** (32 lec · 4hr 34min)
- ☐ **Project 2: Datasource – MySQL Database & SQL Server** (16 lec · 2hr 12min)
- ☐ **Project 3: Datasource – Google BigQuery** (20 lec · 2hr 47min)
- ☐ **Project 4: Datasource – Azure SQL Database** (16 lec · 1hr 58min)

💡 Do Projects 2 + 3 fully (MySQL + BigQuery). Projects 1 + 4 can be watched at 1.5× – still code along. These show the full modern data stack.

### ■ AI TOOLS + GITHUB + PROMPT ENGINEERING

- ☐ **New Power BI Project Using AI Tools** (17 lec · 1hr 47min) – use AI to generate/debug DAX, build a full project with AI assistance
- ☐ **Creating GitHub Account & Uploading Power BI Projects to GitHub** (2 lec · 14min) – course teaches GitHub explicitly here
- ☐ **Prompt Engineering & Generative AI For Data Analytics** (29 lec · 3hr 36min) – AI prompting for SQL, Python, Power BI, data storytelling. Final section of the course.

### 🦋 POST-COURSE CAPSTONE (BUILD AFTER WEEK 7)

#### Python → SQL Server → Power Query → Power BI + DAX → Published → GitHub

*Skills: ALL course tools – Python · Pandas · SQL Server · Power Query · Power BI · DAX · GitHub*  
*README*

- ☐ Dataset: Flipkart/Amazon India sales, 50k+ rows (Kaggle)
- ☐ Step 1 – Python/Pandas: full cleaning pipeline as .py script (not just notebook)
- ☐ Step 2 – SQL Server: load data, 10+ queries (window functions, CTEs, interview patterns)
- ☐ Step 3 – Power Query: connect SQL Server → Power BI, build data model
- ☐ Step 4 – Power BI: 4-page report, 8+ DAX measures, drill-through, publish to Service
- ☐ Step 5 – GitHub: README with architecture diagram, SQL files, screenshots, live PBI link
- ☐ GitHub: 6+ repos with proper READMEs + screenshots pushed
- ☐ LeetCode: ~50 SQL problems done (Easy + some Medium)
- ☐ ☒ **COURSE COMPLETE.** India job search starts immediately – Day 1 of Week 8.



## ■ GOAL 2 – INDIA DA JOB TRACK (Within 3 Months of Course End)

Weeks 8-14 · 3 hrs/day · Apply from Day 1

💡 You have ALL the core skills. The India DA market wants: SQL (strong), Power BI, Python basics, Excel, communication. You have more than most applicants. Start applying in Week 8. Don't wait for "more projects."

### 🔗 INDIA PROJECT 1 – Zomato Restaurant Analysis

*Skills combined: Pandas · Matplotlib · Seaborn · SQL (SQLite) · Tableau or Power BI · GitHub README*

- ☐ Dataset: Zomato restaurants India dataset (Kaggle) – 50k+ rows
- ☐ Python: clean cuisines/ratings columns, handle missing, feature engineer city-level aggregates
- ☐ SQL: 8 queries – top-rated by city, most common cuisines, online vs offline delivery, price range analysis
- ☐ Dashboard: Power BI or Tableau – city comparison, rating distribution, delivery type breakdown
- ☐ GitHub README: headline insight per city + "What would you recommend to a new restaurant owner?"

### 🔗 INDIA PROJECT 2 – IPL Data Analysis (2008-2024)

*Skills combined: Pandas · NumPy · Matplotlib · Seaborn · SQL · Power BI (DAX for player/team KPIs) · GitHub*

- ☐ Dataset: Kaggle IPL complete dataset (matches + deliveries)
- ☐ Python: merge match + delivery data, compute strike rates, economy rates, win margins
- ☐ SQL: top batsmen per season, team head-to-head, toss decision impact on wins
- ☐ Power BI: season-by-season trends, player performance cards, venue analysis
- ☐ LinkedIn post: "I analysed 16 years of IPL data – here's what I found" with 3 charts

### 🔗 INDIA PROJECT 3 – Indian E-Commerce Sales Dashboard

*Skills combined: Python · Pandas · SQL · Excel (pivot tables) · Power BI · DAX · GitHub*

- ☐ Dataset: Amazon India sales / Flipkart sales (Kaggle)
- ☐ Excel: clean + pivot analysis first – this is what interviews test
- ☐ Power BI: Revenue by category, state-wise performance, discount impact on profit
- ☐ DAX: Profit margin %, MoM growth, running total, top 10 products by revenue
- ☐ Add a 1-page executive summary PDF to GitHub (pretend you're presenting to a manager)

## INDIA JOB APPLICATION – Weekly Checklist (Weeks 8-14)

Start applying Week 8, Day 1. Don't wait.

### 📄 RESUME (WEEK 8)

- ☐ 1-page resume (India standard: photo optional, 1 page for freshers)
- ☐ Skills: Python · SQL · Power BI · DAX · Tableau · Excel · Pandas · NumPy · Matplotlib · Seaborn · Git · Snowflake · AWS S3 · Azure · BigQuery
- ☐ Projects section: 3 bullet points each, with numbers (e.g. "Analysed 50k rows...")
- ☐ LinkedIn: update headline → "Aspiring Data Analyst | SQL · Power BI · Python · Tableau"

### 🔗 APPLY EVERY WEEK

- ☐ 10 applications/week: Naukri.com, LinkedIn Jobs, Internshala, Angellist India
- ☐ Search terms: "Data Analyst fresher", "Business Analyst trainee", "SQL Analyst", "Power BI analyst"
- ☐ Target companies: analytics startups, consulting firms (Deloitte/PwC/KPMG India), product companies
- ☐ 5 LinkedIn connections/week with personalised messages to India DA professionals
- ☐ Follow up on every application after 1 week

## ■ GOAL 3 – COMMON SKILLS (Both Europe + Australia Need These)

Learn in parallel with India job search · Adds ~8-10 hrs/skill

### 🌐 COMMON SKILLS – LEARN THESE FIRST

- ☐ **Git Advanced** (4 hrs) – branching strategies (git flow), merge vs rebase, PRs, resolving conflicts, .gitignore, cherry-pick. Resource: Pro Git book (free at git-scm.com)
- ☐ **Scikit-learn ML Basics** (8 hrs) – not ML engineer level, just DA-level: LinearRegression, train/test split, cross-validation, confusion matrix, feature importance. Resource: Scikit-learn official docs + Kaggle Learn ML course (free)
- ☐ **Docker Basics** (6 hrs) – containers, images, Dockerfile, docker-compose, volume mounts. Resource: docs.docker.com Getting Started + Dockerize a Python script
- ☐ **dbt Core** (5 hrs) – SQL transformation tool for modern data stacks. Resource: learn.getdbt.com Fundamentals (free, exactly 5 hrs). Complete the free certification.
- ☐ **Streamlit** (8 hrs) – turn Python scripts into live web apps. Resource: docs.streamlit.io. Build a Streamlit version of one EDA project. Deploy to Streamlit Cloud (free).
- ☐ **LeetCode SQL** – 2 problems/day ongoing. Milestone tracking below.
- ☐ **Cloud Deeper – Pick ONE:** AWS (Glue + Athena) OR Azure (Data Factory + Synapse). Pick based on which track you lean toward. Australia → Azure. Europe → AWS or Azure.

### ⚡ LEETCODE SQL MILESTONES

- ☐ **25 problems done** – Easy all cleared. Move to Medium.
- ☐ **50 problems** – Core SQL solid (JOINS, GROUP BY, HAVING)
- ☐ **75 problems** – Window functions and subqueries fluent
- ☐ **100 problems** – CTEs and complex multi-step queries comfortable
- ☐ **150 problems** – Interview ready for any EU/AU SQL test
- ☐ **200 problems** – Elite. Top 10% of DA applicants.

💡 LeetCode Top SQL 50 study plan is the fastest path. After 50, switch to company-tagged questions for the companies you're applying to.

## ■ GOAL 4 – COUNTRY TRACKS (Both Included Europe + Australia side by side. Common skills first, – Decide Later) then pick your track.

### EU EUROPE TRACK

Missing skills to add beyond common: Linux/Bash · FastAPI/REST APIs · PySpark · Deep Learning basics · Open Source contributions

- ☐ **Linux/Bash** (6 hrs) – navigate filesystem, pipes, grep, awk, sed, cron, ssh, chmod. Resource: OverTheWire Bandit (free, gamified), or Linux Journey (linuxjourney.com)
- ☐ **FastAPI / REST APIs** (8 hrs) – build a simple REST API that serves your ML model or SQL results as JSON. Resource: FastAPI docs (fastapi.tiangolo.com) – the "First Steps" tutorial covers 80% of what DAs need
- ☐ **Scikit-learn ML Basics** (covered in common – include in EU context: regression, classification, feature importance reports)
- ☐ **Deep Learning Basics** (6 hrs) – not DL engineer level. Understand: neural network structure, activation functions, loss functions, when DL vs ML. Resource: fast.ai Practical Deep Learning (free), do Part 1 only
- ☐ **PySpark** (8 hrs) – distributed data processing. Used in EU enterprise data teams heavily. Resource: Databricks Community Edition (free) + PySpark Getting Started docs. Focus: RDDs, DataFrames, groupBy, joins at scale
- ☐ **Open Source Contributions** (ongoing) – fix 2 bugs in a Python data library (pandas-related GitHub issues). Makes your GitHub credible. Start here: goodfirstissue.dev
- ☐ **Docker** (covered in common – EU context: Dockerize your FastAPI endpoint)
- ☐ **Git Advanced** (covered in common – EU context: contribute to open source using proper PR workflow)

### AU AUSTRALIA TRACK

Missing skills to add beyond common: PL-300 Cert · Tableau deeper (LOD) · Azure Data Factory · Looker Studio

- ☐ **PL-300 Certification** ~AUD \$165 – Microsoft Power BI Data Analyst Associate. 45–60 questions, 120 min, pass = 700/1000. Study: Microsoft Learn (free). Practice: ExamTopics.com. Book: Pearson VUE. Study 1 hr/day × 3 weeks before sitting.
- ☐ **Tableau Deeper – LOD Expressions** (6 hrs) – FIXED, INCLUDE, EXCLUDE. LOD is what separates basic from advanced Tableau. Resource: Tableau's own "LOD Expressions" whitepaper + Tableau Public practice workbooks
- ☐ **Azure Data Factory** (5 hrs) – Microsoft's cloud ETL tool. Dominant in AU government and enterprise (most AU orgs run Microsoft stack). Build one pipeline: Blob Storage → Azure SQL using ADF. Resource: Microsoft Learn ADF Getting Started (free, ~3 hrs) + 2 hrs hands-on
- ☐ **Looker Studio** (4 hrs) – Google's free BI tool. Some AU companies use Google Workspace. 1 weekend. Connect to BigQuery or a Google Sheet. Publish one dashboard. Push screenshots to GitHub.
- ☐ **AWS Deeper: Glue + Athena** (6 hrs – from common) – AU mining and tech sector uses AWS heavily. Glue = managed ETL. Athena = query S3 with SQL directly. Resource: AWS free tier + AWS Glue Getting Started (2 hrs) + Athena tutorial (2 hrs) + small pipeline (2 hrs)
- ☐ **Databricks Intro** (6 hrs) – rapidly growing in AU enterprise. Community Edition is free. Complete the SQL Analytics notebook tutorial. Push notebook to GitHub.

### EU EUROPE PROJECTS

### AU AUSTRALIA PROJECTS

**EU PROJECT 1 – FastAPI ML Serving Endpoint**

*Skills: Scikit-learn · FastAPI · Docker · Git · GitHub Actions*

- ☐ Train a simple Scikit-learn model (flight price prediction or wine quality)
- ☐ Serve it as a REST API endpoint using FastAPI (POST /predict)
- ☐ Dockerize the app – include Dockerfile
- ☐ Set up GitHub Actions to run tests on every push
- ☐ GitHub README: architecture diagram, API docs, how to run locally

**EU PROJECT 2 – PySpark + dbt + Power BI Pipeline**

*Skills: PySpark · dbt · Snowflake · Power BI · DAX · GitHub*

- ☐ Use Databricks Community (free) – process a 1M+ row dataset with PySpark
- ☐ Export aggregated data → run through dbt models in Snowflake
- ☐ Connect Snowflake → Power BI for final dashboard
- ☐ This replicates a real EU enterprise data stack

**EU PROJECT 3 – Open Source Contribution**

*Skills: Git Advanced · GitHub · Python · Code review workflow*

- ☐ Find 2 "good first issue" bugs on pandas, matplotlib, or dbt GitHub repos
- ☐ Fork → fix → test → PR with proper description
- ☐ Even if PR isn't merged, it shows process knowledge – list on resume

**AU PROJECT 1 – PL-300 Power BI Showcase**

*Skills: Power BI · DAX · Power Query · Azure Data Factory · dbt · Snowflake*

- ☐ Use data.sa.gov.au or data.gov.au (Australian local dataset)
- ☐ Build the pipeline: ADF → Azure SQL → Power BI
- ☐ 4-page Power BI report, 8+ DAX measures, RLS, drill-through
- ☐ Publish to Power BI Service – get a shareable public link
- ☐ Write a LinkedIn post: "I analysed [AU topic] using Power BI – here's what I found"

**AU PROJECT 2 – Tableau LOD Deep Dive Dashboard**

*Skills: Tableau · LOD Expressions (FIXED/INCLUDE/EXCLUDE) · Tableau Prep Builder*

- ☐ Dataset: Superstore or Australian health/property open data
- ☐ Use Tableau Prep Builder to clean and shape data
- ☐ Build 3 LOD calculations: FIXED (YTD total), INCLUDE (avg per customer), EXCLUDE (comparison without dimension)
- ☐ Publish to Tableau Public – shareable URL on resume

**AU PROJECT 3 – Streamlit + Docker Full-Stack App**

*Skills: Python · Streamlit · Docker · AWS S3 or Azure Blob · GitHub*

- ☐ Build an interactive Streamlit dashboard on Australian open data (health/housing/transport)
- ☐ Load data from AWS S3 or Azure Blob Storage (cloud-native data source)
- ☐ Dockerize the app – include Dockerfile + docker-compose.yml
- ☐ Deploy to Streamlit Community Cloud – get a public live URL for resume

## ■ COMPLETE SKILLS REFERENCE – Status Tracker

Tick each skill as you become interview-ready in it

### ■ FROM COURSE

- ☐ Python (syntax → functions → OOP)
- ☐ Pandas + NumPy
- ☐ Matplotlib + Seaborn
- ☐ SQL (all topics + assignments)
- ☐ Statistics + Feature Engineering
- ☐ Power BI + Power Query
- ☐ DAX (all 6 lecture groups)
- ☐ Excel (functions + dashboard)
- ☐ Tableau (all + Prep Builder)
- ☐ Snowflake basics
- ☐ AWS S3 basics
- ☐ Azure basics
- ☐ BigQuery basics
- ☐ AI/Prompt Engineering tools

### ■ COMMON (EU+AU)

- ☐ Git + GitHub (basic)
- ☐ Git Advanced (branching, PRs)
- ☐ Scikit-learn ML basics
- ☐ Docker basics
- ☐ dbt Core (+ free cert)
- ☐ Streamlit (+ deployed app)
- ☐ LeetCode SQL 50+ problems
- ☐ LeetCode SQL 100+ problems
- ☐ LeetCode SQL 150+ problems
- ☐ Cloud deeper: AWS or Azure

### ■ COUNTRY-SPECIFIC

#### EU Europe Only

- ☐ Linux/Bash scripting
- ☐ FastAPI / REST APIs
- ☐ Deep Learning basics (fast.ai)
- ☐ PySpark (Databricks)
- ☐ Open Source contribution (2 PRs)

#### AU Australia Only

- ☒ PL-300 Certification
- ☐ Tableau LOD Expressions
- ☐ Azure Data Factory
- ☐ Looker Studio
- ☐ Databricks Intro

## ■ INTERVIEW PREP – STAR STORIES FROM YOUR PROJECTS Prepare these before your first interview

- ☐ **"Tell me about a time you worked with messy data"** → Google Playstore EDA or Zomato project. Walk through: what was messy, how you handled it, what you found.
- ☐ **"Tell me about a time you delivered a business insight"** → Post-Week 7 Capstone or India E-Commerce project. State 1 specific finding with a number.
- ☐ **"What SQL skills do you have?"** → Mention: JOINS, CTEs, Window Functions, Stored Procedures. Then say: "I also regularly practise on LeetCode – [X problems done]."
- ☐ **"Describe a complex technical problem you solved"** → Post-Week 5 end-to-end project or EU/AU pipeline project. Architecture: Python → SQL → Power BI.
- ☐ **"What's your best project?"** → Post-Week 7 Capstone. Walk through every tool used and one business insight it revealed.
- ☐ **Live SQL test prep** – practice writing queries out loud. Explain your thinking as you type. Use Pramp.com (free mock interviews) or ask a friend to question you.

### ALL PROJECTS – QUICK REFERENCE (Tick when complete + pushed to GitHub)

Every project needs: proper README · screenshots · pushed code · clear skills-used section

- ☐ **W5** Red Wine EDA
- ☐ **W5** Flight Price EDA
- ☐ **W5** Google Playstore EDA (1+2)
- ☐ **W6** Power BI Sales Data Analysis
- ☐ **W6** Power BI Insurance Data Analysis
- ☐ **W6** Power BI UPI Transactions Analysis
- ☐ **W6** SQL + Tableau Student Depression
- ☐ **W6** Excel Dashboard 1 + Dashboard 2
- ☐ **W7** 4x Cloud End-to-End PBI Projects
- ☐ **Post-W7** Capstone (Python→SQL→PBI)
- ☐ **India** Zomato Restaurant Analysis
- ☐ **India** IPL Data Analysis (2008–24)
- ☐ **India** E-Commerce Sales Dashboard
- ☐ **EU** FastAPI ML Serving Endpoint
- ☐ **EU** PySpark + dbt + Power BI Pipeline
- ☐ **EU** Open Source Contribution (2 PRs)

☐ **AU** PL-300 Power BI Showcase (AU data)☐ **AU** Tableau LOD Deep Dive Dashboard☐ **AU** Streamlit + Docker Full-Stack App**🚫 WHAT TO SKIP / SKIM (save 10+ hours):**

**SKIP:** Deep stats theory (ANOVA derivation, chi-square calculations) · OOP advanced (decorators, metaclasses) · Tableau Server admin sections · Snowflake admin/billing sections · AWS advanced networking · Azure DevOps pipelines

**SKIM (20 min max each):** Z-test, T-test, ANOVA, Chi-Square (know what they are, not how to calculate by hand) · Excel macros (know they exist) · Power BI custom visuals marketplace (know it exists) · Tableau Server publish workflows (you use Tableau Public – free)

**PRINT THIS. STICK IT ON YOUR WALL. TICK EVERY BOX.**